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Falsification-First Decipherment: A Decontaminated Inference Framework for Testing Undeciphered-Script Claims, with Linear A as a Worked Null

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Anonymized submission copy — author, repository, and artifact-deposit identifiers are withheld for double-blind review. The method name "logos" is retained as the system name; a public, non-anonymized version with full provenance is archived separately.

Abstract

Decipherment research has a false-positive problem and no shared instrument for measuring it: a small corpus, an unknown language, and a vast space of candidate readings let an analyst "translate" almost anything, with internal consistency standing in for confirmation. The canonical absurdity — "proving" English is Semitic by cherry-picking root matches — recurs whenever the many comparisons a reading was selected from go uncounted. The field's flagship survey enumerates fifteen challenges — primarily script, corpus, language, resource, and comparison-data constraints; none address the orthogonal inference-control layer of overfitting, multiple testing, false-positive control, or contamination. We frame that methodological axis, modestly, as a candidate sixteenth challenge and build machinery for it: a falsification-first, decontaminated inference framework that treats every positive as guilty until proven innocent. Four defences are enforced — a fabricated-language canary supplying a false-positive floor, multiple-comparison deflation over signs, roots, and families tried, a literature index separating published from literature-unseen signs, and pre-registration with a held-out gate graded mechanically rather than by the proposing model. This is a methods paper; we assert no decipherment. Linear A is worked as the honest null: the binding constraint is identifiability, not corpus size or effort. We report a body of pre-registered nulls with their borderline cases named, and analyse the June-2026 *301 "cracked" claim as a worked failure case. The artifact is open and archived ([repository URL withheld for review]; [artifact DOI withheld for review]).

1. Introduction

Decipherment has a false-positive problem, and the field has no agreed instrument for measuring it. The history of undeciphered-script research is, in large part, a history of self-deception: a tiny corpus, an unknown language, and a large space of candidate readings let a determined analyst "translate" almost anything, with internal consistency on the cherry-picked set standing in for confirmation. The mechanism is not any single bad match but an unpaid multiplicity: with enough candidate roots and a small enough corpus, a pile of individually-plausible matches is the *expected* outcome under pure chance, so "each individual comparison looks suggestive [while] the conclusion is nonsense". The canonical demonstration is deliberately absurd — one can "prove" that *English is a Semitic language* by matching a handful of English words to Proto-Semitic roots while quietly ignoring the far larger number that do not match. The same move, applied to a real undeciphered script, has produced a long line of decipherment claims that no one can either confirm or refute.

That the discipline lacks a shared standard here is not our editorializing but the field's own assessment. The flagship computational-decipherment survey — Braović et al.'s (2024) systematic review of Bronze Age Aegean and Cypriot scripts — enumerates fifteen "major challenges," from unknown writing system and unknown language to small dataset, incomplete vocabulary, allographs, and unavailable parallel data (Braović et al. 2024). Those fifteen are primarily properties of the *object and its data* — the script, the language, the surviving corpus, and the resource/comparison-data conditions (the list includes resource and hardware items too). What governs whether a *claimed reading* is believable is a different kind of thing: a property of the inference methodology and its epistemics — overfitting, multiple testing, false-positive control, and contamination. We therefore frame this axis, once and modestly, as a candidate **sixteenth challenge** — a deliberate rhetorical foil to the survey's fifteen, not a claim to a new taxonomic slot or to sole discovery. It is *under-instrumented* rather than unrecognised: Packard's (1974) nine deliberately fabricated decipherments already exercised exactly this false-positive logic as a control, and the selective-inference machinery we import (Bailey & López de Prado 2014) is mature elsewhere — our contribution is to *consolidate and operationalize* that axis for decipherment, not to invent it. It is the methodological axis the object-level taxonomy was never meant to cover, and it is distinct from the survey's own acknowledged gap — that on scoring results "in many cases there is nothing to compare them against, and their evaluation remains an open problem" (Braović et al. 2024, Sec. 6.3, p. 746). That is an *evaluation-target* problem (what to measure against); the false-positive / multiplicity problem (whether a fitted reading is being mistaken for a real one) is separable from it, and it is the latter this paper instruments.

Contribution. We build the machinery for that axis: a falsification-first, decontaminated inference framework for testing undeciphered-script decipherment claims (Braović et al. 2024). The harness treats every positive as guilty until proven innocent and enforces four concrete defences: a fabricated-language canary that supplies a **generator-conditional negative-control distribution** (a false-positive floor conditional on the fabricated-language generator — here a first-order sign-bigram Markov model; a broader control ensemble spanning unigram, root-template, and real-language generators is disclosed as future work, not built here); multiple-comparison deflation for the number of signs, roots, and families a search was free to try; a literature index that partitions published (`L_known`) from literature-unseen (`L_virgin`) signs so that reproducing a known value cannot masquerade as discovery; and pre-registration with a held-out gate whose verdict is computed mechanically, never by the model that proposed the reading. The evaluation gap gets a concrete, if power-limited, answer: the recurring libation *formula* — roughly 50 inscribed objects spread across peak sanctuaries and caves at more than five findspots — supports a natural leave-one-site-out cross-validation graded from held-out data. We are explicit that with so few formula sites this cross-validation has low statistical power, and we report it as such rather than as a decisive test (Braović et al. 2024).

This is a methods paper, and Linear A is its worked null. We do not claim a decipherment. If one is attainable on present evidence, the framework is built to recognize it; when nothing survives, the calibrated null — what the corpus can and cannot support — is itself the deliverable. The working conclusion on current data is that the binding constraint on Linear A is **identifiability**, not corpus size or effort: there is no known related language to map onto, the multiple-testing burden of candidate readings goes uncorrected, and the sign inventory is mixed and partly lossy. A lossy, non-injective sign-to-sound channel need not admit a decipherment at any corpus size, so we make no claim that "there is enough text to pin a reading"; the rigorous demonstration of *that* constraint is the result the field can use. The June-2026 AI-assisted "Linear A cracked" claim — reading the `\*301`-anchored libation verb as an extinct Semitic form — serves throughout as a worked example of exactly these failure modes, analysed from its public, archived presentation as a claim, not a person, and neither reproducible nor held-out validated. The artifact is open and archived ([repository URL withheld for review]; [artifact DOI withheld for review]).

Roadmap. Section 2 situates logos against the Braović et al. taxonomy and the computational-decipherment lineage, including the cognate-ID baseline. Section 3 specifies the harness — canary, deflation, literature index, and the pre-registration/held-out gate. Section 4 develops the information-floor diagnostic and states its identifiability limits. Section 5 works the June-2026 `\*301` claim as a case study of the failure modes. Sections 6–8 report the pre-registered nulls: morphology (§6); metrology, phonology, and cross-script signals (§7); and contamination and sufficiency probes (§8). Section 9 draws the discussion, limitations, and conclusion.

2. Related work

2.1 Position relative to the survey taxonomy

The reference frame for computational work on the Bronze Age Aegean scripts is the systematic review of Braović, Krstinić, Štula and Ivanda (2024, *Computational Linguistics* 50(2):725–779), the field's first survey focused specifically on Aegean and Cypriot scripts (its only comparable predecessor being Ferrara and Tamburini 2022); the broader machine-learning-for-ancient-languages landscape is surveyed by Sommerschield et al. (2023, *Computational Linguistics* 49(3)). Braović et al.'s Section 4 enumerates fifteen "major challenges" of computational decipherment — from unknown writing system, reading direction and language, through small dataset, allographs and incomplete vocabulary, to unavailable parallel data and hardware. None of the fifteen concerns overfitting, multiple testing, false-positive control, contamination, or the problem of distinguishing a *fitted* decipherment from a *real* one. The same gap runs through Ferrara and Tamburini's (2022, *Lingue e Linguaggio* XXI.2:239–259) predecessor survey: its §3.5 reviews the very cognate-matching lineage logos builds on — Snyder et al. (2010) on Ugaritic↔Hebrew (the 29/30-sign benchmark our canary calibrates against), Berg-Kirkpatrick and Klein (2011), and Luo et al. (2019, 2021) — and frames Linear A, with Gelb and Whiting's typology, as a Type III script (unknown script *and* unknown language) whose "necessary validation" is "altogether impossible," yet nowhere treats how a proposed match is to be *deflated* for the multiplicity it was selected from or separated from a false positive. Two field surveys, the same unfilled methodological axis. We do not score this against either as an oversight: their challenges are properties of the object and its data — an unknown language, a small dataset, allographic variation, missing parallel text — whereas the axis logos foregrounds is a property of inference *methodology* and epistemics, a deliberately different kind of thing. The false-positive / multiplicity axis is therefore named-but-unfilled rather than absent, and logos supplies the machinery to police it: preregistration, a decontaminated derivation/held-out partition, multiplicity deflation, and a graduation gate computed mechanically rather than by the model that proposed the reading.

Braović et al. do gesture at a related but separable difficulty: evaluation. Because computational analyses of an undeciphered script have "nothing to compare them against," they conclude (Sec. 6.3) that "their evaluation remains an open problem." That is a distinct problem from overfitting and multiplicity control, and logos addresses it separately. Its held-out design is a concrete response — the libation formula recurs across a small set of peak-sanctuary and cave findspots, permitting a leave-one-site-out cross-validation adjudicated mechanically by `verdict.py` — though we flag at the outset that so few formula findspots give this particular test low statistical power (quantified in Section 8). We locate our components in the survey's own method taxonomy (Sec. 7.1) and, where prior art exists, defer to it rather than claim novelty.

2.2 The current philological synthesis

Salgarella (2025), *Writing in Bronze Age Crete: "Minoan" Linear A* (Cambridge Elements), by the author of the SigLA database, is the most current authoritative one-author overview. It concludes that Minoan is a language **isolate** — it "does not belong in any known language family" (the families used for comparison being Indo-European, Semitic and Afro-Asiatic) — typologically agglutinative with a provisional Verb–Subject–Object order, and it records that "no bilingual text has so far been found" in a corpus of roughly 2,500 inscriptions. Crucially, Salgarella ranks the etymological/comparative method below combinatorial-contextual analysis and names **digital/statistical analysis as "the next desideratum in the field."** logos positions itself as exactly that desideratum — but under discipline, treating the isolate verdict as the binding *identifiability* constraint rather than a target to be argued away.

2.3 Prior computational attempts and the search trap

The seminal automatic-decipherment result is Luo, Cao and Barzilay (2019), which recovers a character mapping from a lost script to a *known related* language via neural minimum-cost flow. It auto-deciphered Ugaritic→Hebrew (+5.5% over prior state of the art) and translated 67.3% of Linear B→Greek cognates, but did not address Linear A and would fail there by construction: with no known cognate plaintext, the objective has nothing to optimise toward. This cognate-language precondition is the empirical form of our identifiability argument. The cognate-ID baseline logos actually runs is *not* a reimplementation of Luo's neural flow: it is a published matcher, Tamburini (2025)'s CSA_OptMatcher (coupled simulated annealing), which reports the same Luo 2019 accuracy metric. We adopt it as the non-neural baseline against which any Linear A cognate claim must clear.

Against that disciplined baseline stand the in-domain instances of the search trap: the dictionary-fishing programmes of Min Eu, Xu and Cacciafoco (2019) and Loh and Perono Cacciafoco (2020), the latter explicitly a "brute force attack" on dictionaries. These are the concrete Linear-A cases the multiplicity correction is built to deflate. On the visual side, cross-script comparison is *not* our novelty: Daggumati and Revesz (2019, 2023) already applied CNN+SVM cross-script similarity, and Corazza et al. (2022, "Sign2Vec") clustered signs unsupervised; our defensible contribution is only the controlled image-versus-sequence *asymmetry* under one harness.

2.4 The discipline lineage

Two older strands supply the honesty machinery. Packard (1974), *Minoan Linear A*, constructed nine deliberately fictitious decipherments as a null control — and the genuine Ventris-value transfer beat them only about 2:1 — anticipating our permutation-null canary. The statistical lineage is selective inference and deflated significance (Bailey and López de Prado 2014), the correction for the number of hypotheses × signs × roots × families tried that turns a raw match-rate into a defensible one; its transfer from the finance/backtest setting to script decipherment is specified in Section 3. logos's empty cell is the intersection none of these occupy: a falsification-first, decontaminated inference framework that couples a published cognate-ID matcher (Tamburini 2025, reporting the Luo 2019 metric) with preregistration, multiplicity deflation and mechanical held-out verdicts, released openly ([repository URL withheld for review]; [artifact DOI withheld for review]), with Linear A worked as the honest null rather than a claimed crack.

3. Methods: the discipline harness

The harness is a comparison-layer pipeline that a reader can reimplement. A candidate reading enters as a *proposal* and is subjected, in fixed order, to five gates: pre-registration, deflation, decontamination, held-out graduation, and a cross-cutting rule that every gate is scored from persisted artifacts rather than from any model's narration of its own success. Model-derived signals — LLM theses, learned embeddings, cognate-matcher assignments — enter capped at confidence ≤ 0.75, structurally unable to *decide* a verdict; only mechanically-verified held-out reads rank higher. The design targets the false-positive / multiplicity / contamination axis, which Braović et al. (2024)'s fifteen-challenge taxonomy names but leaves unfilled. Their fifteen challenges are properties of the object and its data — an unknown language, a small corpus, allographs; the axis logos adds is a property of inference *methodology* and epistemics, deliberately a different kind of thing. (Their §6.3 note that "evaluation remains an open problem" gestures at evaluation, a distinct problem from overfitting and multiplicity; the two are kept separable here.)

3.1 Pre-registration

Before any model is built or run, the predictions are frozen. The morphology pre-registration fixes, for each hypothesis, the claimed affix, its source, the falsifiable test, and the acceptance threshold — with the git commit of the file serving as the timestamp, so the model cannot be tuned to them after the fact. Each pre-registered affix (drawn from Thomas 2020 and Duhoux/Valério) must recur on ≥2 distinct stems across independent inscriptions, above a within-form permutation null, and survive multiplicity correction at a target of $p < 0.01$ corrected. Outcomes are enumerated in advance — CONFIRM, EXTEND, REFUTE/NULL, or NO POWER — so that a negative is a publishable result, not a failed experiment. Any change is a new dated pre-registration, never an edit.

3.2 Deflation

Internal fit on the derivation set is treated as no evidence. Every match rate is deflated for everything tried: `n_hypotheses × n_signs × n_roots`. The nulls are pipeline-level — a frequency-banded permutation null (after Packard 1974) and an Euler-γ order statistic (`expected_max_order_stat(μ₀, σ₀, N)`) that gives the bar a maximum over N effectively-independent candidates should clear by chance. Multiplicity is instrumented, not hand-passed: a deduplicating `SearchLog` reports the exact distinct-candidate count `n_eff` a scanner produces (invariant 12). The strong structural statistic, `S_morph`, earns credit only through a productivity gate — an affix must attach to ≥2 distinct residual stems, so a single formula word copied across inscriptions scores zero. After that fix, its measured false-positive rate on within-form-shuffled corpora was **3.0% (6/200; Wilson 95% CI [1.4, 6.4]%)**, consistent with the nominal one-sided $z \geq 2$ rate ($\approx$ 2.3%) within sampling error. We flag this explicitly as an **indicative calibration check, not the operative gate**: a 6/200 estimate is too coarse to certify an $\alpha = 0.01$ threshold on its own, and it is reported only to show the detector is not grossly mis-calibrated at the nominal level. The graduation gate does not rest on it — it applies the order-statistic bar over the far larger permutation and candidate counts the deflation machinery tracks (§3). Power was **100% (50/50)** at a single strong planted-affix strength — a saturated point, not a full sensitivity curve (contrast the phonology power curve, §7).

3.3 Decontamination

A frontier model has read Gordon, Best, and Di Mino, so a proposed correspondence that merely reproduces a *published* value is shared-source contamination, not independent evidence. Three mechanisms address this.

The literature index records published Linear A sign-value and correspondence claims; any proposal matching the index is tagged `literature_match` and quarantined — it may be tested but can never count as discovery. The index partitions the sign inventory into `L_known` (referenced by ≥1 published proposal) and `L_virgin` (untouched). Discovery claims may rest only on `L_virgin` held-out success, since regurgitation can only return what is in the literature. The seed is deliberately conservative — a sign wrongly omitted is wrongly granted `L_virgin` status, the dangerous direction — and every disputed entry is indexed to *catch* regurgitation, never as an accepted reading. Population is adversarial: of the seeded West-Semitic lexical proposals, five are Gordon's equations and a-sa-sa-ra-me = "oh Asherah!" is Best (1981), not Gordon; candidates failing independent corroboration against Rendsburg (1996) (`qa-pa` = kappu, a Semitic `ki-ro`) were rejected rather than indexed. Generated from the seed (invariant 12), the index currently holds **63 published claims across 62 indexed entries (55 single signs)** — a deliberately conservative, non-exhaustive seed, not yet the full §C.1 literature census.

The L_fake canary is a phonotactically-plausible invented lexicon, calibrated to a real candidate language's phoneme frequencies, root-template structure, and lexicon size, but never published and never in any training corpus. Running the full pipeline against it yields an empirical false-positive floor: any signal it produces is definitionally spurious, and memorisation cannot be invoked because there is nothing to memorise. A real candidate's match must clear the L_fake score distribution by the corrected margin. Its own divergence from the calibration targets is reported, never minimised; with a Semitic root-template sampler and an external Hebrew (ETCBC/bhsa) reject set, the root-template total-variation distance fell from ~0.84 to ~0.33 end-to-end (~0.23 standalone) on the Ugaritic↔Hebrew gold, and any residual real-lexicon collision is measured and published rather than asserted to be zero.

LLM-ablation compares the pipeline with and without the model's proposals to isolate contamination — the remaining gate on the contamination number, matching a model's cognate glosses against the indexed lexical claims.

3.4 The held-out graduation gate

A reading graduates only mechanically, from held-out data, as a conjunction of pre-registered clauses computed by `scripts/verdict.py` — never by the model that proposed the reading. The operative deflation clause is an **order-statistic bar**: the observed held-out statistic must exceed the *expected maximum* of the multiplicity-null taken over the number of trials the search was free to run (the E[max] over `n_trials` deflation). This is a direct correction for selection across the hypothesis space, and it fails closed on a degenerate null (zero variance ⇒ no graduation). The Deflated-Sharpe Ratio (Bailey & López de Prado 2014) — a probability in [0, 1], the confidence that the true deflated performance exceeds its multiplicity-adjusted benchmark — is computed and **reported** alongside; after pre-submission review we made the order-statistic bar the *operative* clause rather than a fixed DSR ≥ 0.95 cutoff, because the expected-max bar needs no normal approximation and fails closed on degenerate input. The match-rate the deflation is applied to is supplied by a cognate-ID baseline that reports the Luo et al. (2019) accuracy metric; the instance we run is Tamburini's CSA_OptMatcher (coupled simulated annealing, §2.3–2.4, reproduced in the artifact), so the gate does not depend on any single citation. A second pre-registered clause, **`generalizes_to_virgin`**, requires the reading to recover literature-unseen (`L_virgin`) signs above a pre-committed threshold and on a minimum number of distinct such signs — so reproducing already-published values cannot by itself graduate a claim, and the clause fails closed when no threshold or no supported virgin sign is present. The L_fake corpus must *fail* as a null, its corrected margin must be cleared, and the observed statistic must beat the order-statistic bar over the Packard/random-lexeme multiplicity null (the within-inscription sign-order shuffle is a null in the morphology module, §6, not a gate clause). The minimum-description-length check (k ≤ U_floor, §4) was, after the same review, **removed from the gate and is now reported as a diagnostic rather than enforced as a graduation clause**: clearing it does not certify identifiability (it is a necessary-precondition / toy-model diagnostic, §4, §9), and as a hard clause it could only be evaluated against a free-parameter default that risked passing degenerate inputs. U_floor is still displayed next to every claim, and "parameters > information ⇒ underdetermined" is still stated (invariant 7) — it simply no longer sits inside the AND. These are *gate parameters* — pre-registered choices for this corpus, not universal constants. Validation is leave-one-site-out, not ordinary *k*-fold, because the libation formula induces formulaic dependence across inscriptions. The gate is AND-monotone — clauses can only make it stricter — so hardening never lets a previously-failing reading graduate.

3.5 Grade from persisted artifacts

The invariant across every gate: verdicts are computed from persisted rows, never from a model's account of its own outcome. The verdict writer is grep-clean of any model call and is the sole writer of the verdicts table. This discipline caught confounds inside logos's own builds — a `S_morph` statistic that conflated repetition with morphology, a verdict clause that was a tautology and could never fire False, and a search-log tripwire that flagged honest multi-sign logs. Each was fixed and locked by regression tests before commit, illustrating the harness turned on itself. The artifact is public and archived (repository [repository URL withheld for review]; [artifact DOI withheld for review]).

4. The information floor

Before any decipherment method is graded, the corpus must be sized against what it can identify. Two quantities matter: how much text exists, and how many free parameters a reading spends against it. Confusing the first for the second is the origin of the "too little text" folk claim — and, we argue, of its equally unearned inverse, the optimism that a big-enough corpus makes a reading identifiable.

The corpus. The current authoritative synthesis (Salgarella 2025) gives an envelope of roughly 2,500 total Linear A finds, of which about 1,400 are readable inscriptions, drawn from 56 findplaces spanning ca. 1800–1450 BCE (Middle Minoan II–Late Minoan IB) but overwhelmingly concentrated in the single LM IB destruction horizon (Salgarella 2025 §1–3, §6). That near-single horizon is why a site-held-out split, not a chronological one, is the only realistic partition. No bilingual text has ever been found (Salgarella 2025 §8).

Which denominator? The recurring sign-count confusion resolves into four distinct denominators, each of which must be labelled where it is used (Salgarella 2025 §1–3): **97** (Braović/Tan token-frequent syllabograms), **~150** (full syllabary estimate), **259** (the logos working repertoire, conflating syllabograms, subscripts, and logograms), and **~350** (total including ideograms; GORILA catalogues 390 signs). The channel through which structure is observed is a *sign-occurrence* count, not a document count or a sign inventory (invariant 7), and two occurrence counts must be kept distinct. The broader **~7,400 sign occurrences** across the whole corpus (Schoep 2002; Salgarella 2025 §1–3) counts every sign type — syllabograms, logograms, and numerals alike. The narrower **N ≈ 5,792 parsed syllable-signs** is the subset that the unicity computation below actually consumes, after logograms and numerals are stripped. The two are different channel sizes and are used in different places; the ~7,400 figure never enters the unicity arithmetic. Two further properties tighten the budget: Linear A words are short and formulaic (the six-position libation-formula template; Salgarella 2025 §7), and a single language/horizon dominates.

The unicity argument, and its limit. On the logos-normalized corpus (1,341 inscriptions, N ≈ 5,792 parsed syllable-signs, V = 259), Shannon's unicity-distance criterion yields a measured range of **U ≈ 204–415 signs** across the whole plausible (V, P) parameter space, everywhere far below N. **We retain this result only as a toy-model precondition diagnostic, and explicitly withdraw it as any sufficiency or refutation claim.** It does *not* show there is enough text to pin a decipherment, and we do not argue that. Unicity distance presupposes a *known plaintext-language oracle*; because we estimate the redundancy D from the ciphertext itself, that oracle collapses into Linear A up to relabelling, so the computation demonstrates only measurable recurrent structure, not that any reading is identifiable. The precondition it checks is necessary, not sufficient. And it can fail to exist entirely: if Linear A underspells like Linear B, the sign→sound channel is lossy and non-injective, and such a channel may possess no unicity distance at all. Corpus size, in short, cannot settle the question either way.

What actually binds. The binding constraint is *identifiability*, not corpus size (Salgarella 2025 §8). It has three parts: (i) the **absence of a known cognate language** — the map has a value-space but no anchored target (Luo 2019's limit); (ii) the **multiple-testing / search trap** — that a unique consistent map may exist in principle neither guarantees an algorithm finds it nor stops a cherry-picked false positive across the parameters tried (invariant 8); and (iii) a **mixed inventory** of syllabograms, logograms, and numerals that a real reading must first separate. These, not the count of surviving signs, are what a decipherment must overcome. Accordingly, when a candidate lexicon grows without bound on a fixed occurrence budget — free parameters exceeding the floor — the reading is underdetermined, and the framework must say so.

5. A qualitative audit of the *301 claim

The framework of Section 3 is abstract until it meets a concrete claim. In June 2026 a widely circulated "Linear A cracked" announcement (Di Mino 2026) argued that Linear A records an extinct Semitic language, reviving Gordon's (1957, 1966) hypothesis. Its load-bearing anchor is a single reading: the recurring peak-sanctuary libation formula word A-TA-I-*301-WA-JA, in which the Linear-A-only sign *301 is *itself* assigned a /na/-initial value, so that the verb root reads as West-Semitic N-W-Y "to dwell" (*nawaya*) and is thereafter matched to later

Hebrew liturgy. We analyse this **as a claim, on its public evidence** — not the person — from its public and archived presentation; our reference copy of the claim and its table is the archived record carried in the artifact (version and details withheld for review), and the repository ([repository URL withheld for review]; [artifact DOI withheld for review]) carries the machinery that indexes it. It is a clean, self-contained instance of exactly the failure modes the gate is built to catch. The refutation is three-part and rests, wherever possible, on the claim's **own published table** and on primary sources. We flag the meta-point explicitly: this refutation is a literature-and-primary-source argument, not itself a logos held-out result — it critiques the claim on shared evidentiary ground rather than adjudicating it mechanically through our own gate. That gap is the sharpest limitation of the present paper, and we name it: the graduation gate and the L_fake canary are validated as instruments (on unit tests and on the known-answer Ugaritic↔Hebrew surrogate) but have not been *run on this claim*. The natural demonstration — scoring Gordon's West-Semitic equations and the Di Mino lexicon against the L_fake false-positive floor and the order-statistic bar, so the reading either clears the machinery or fails closed against it — is directly runnable with the components of §3 and is the priority of the planned revision (docs/findings/2026-07-01-referee-simulation). Until then, "falsification-first" is demonstrated on synthetic positives and argued (not yet mechanically adjudicated) on this real one. Every philological point in the three-part refutation below is drawn from the published literature (Davis 2013, Duhoux 1992, Thomas 2020, Salgarella 2025, Steele & Meissner 2017) or read off the claim's own table — none is advanced in logos's own voice, and the paper introduces no new sign reading. The refutation therefore deploys the field's own readings against the June claim; that it has not been independently vetted by Aegean epigraphers is a disclosed limitation, not a standard the work asserts it meets.

(a) The segmentation is not novel. Root i-*301 plus affixation is the reading of Davis (2013); an independent segmentation by Thomas (2020) reaches the same core and is tested head-to-head against it (the current synthesis, Salgarella 2025, does not itself adopt Thomas). On the Davis and Thomas segmentations, the affix-stacking is evidenced by multi-attestation forms — ta-na-i-*301-u-ti-nu (IO Za 6) and a-ta-i-*301-de-ka (ZA Zb 3) — that vary the material around a stable i-*301 core. (These sigla are confirmed against Davis 2013: `TL Za 1`, `IO Za 6`, and `ZA Zb 3`.) The claim re-segments a structure already established in the literature. Under the decontamination layer this matters procedurally, not just historically: a correspondence that reproduces a published segmentation is shared-source contamination, tagged `literature_match` and barred from counting as discovery (logos v0.1.0, §C.1).

(b) The gloss is contested and weaker. The mainstream value of the verb is "give/dedicate" — Duhoux's (1992) gloss, adopted by Davis (2013) as a *structural* reading that assigns *301 **no phonetic value at all**, and endorsed by the current synthesis (Salgarella 2025, §7.2). "Dwell" is a semantically distant, unmotivated substitution. The point is not that "give" is certainly right, but that a *structural* reading needing no sound-value for *301 already accounts for the data the phonetic claim is invoked to explain.

(c) The primary descriptions of this form's morphology do not support the Semitic label. The claim's own table segments the verb as prefix-stacking and agglutinative — [prefix A "I"] [stem-morpheme TA] [stem-vowel I] [root *301]. We deliberately do **not** rest on an *a-priori* typological contradiction here, and we retract the earlier draft's claim that a prefix-stacked verb is "internally incoherent" with a Semitic reading: Semitic verbal morphology includes productive prefix conjugations, so a prefix on the verb is not by itself un-Semitic. The point is empirical, not typological, and it comes from the primary literature. Davis (2013, p. 38 n. 13) characterises i-*301's morphology as "neither IE nor Afroasiatic" — and Semitic *is* Afroasiatic, so a decade earlier the primary description already found **no Afroasiatic morphological signature** in this very form. (We rest on what the footnote states — the absence of an Afroasiatic signature — not on a stronger "ruled out" than a hedged footnote can bear.) Independently, Salgarella (2025, §8) concludes Minoan is agglutinative (after Duhoux 1978) and a language isolate. The claim's own segmentation is thus consistent with the isolate typology, while the specific Semitic assignment finds no morphological support in the primary descriptions of the form.

The free-parameter receipt. The decisive point for identifiability is that *301 has no cross-script anchor: in the claim's own table its AB-number column is "—" and its Linear B correspondence "(none)". Even the standard practice of reading Linear A homomorphs with their Linear B values is a backward projection to be handled with care (Steele & Meissner 2017; already cautioned by Duhoux 1989, apud Davis 2013, p. 38 n. 11) — and an A-only sign such as *301 has no homomorph to project from at all, so nothing cross-script constrains its value. The single sign on which the entire decipherment pivots is thus a **free parameter** — its value was assigned to fit the desired root, not constrained by any independent evidence. This is precisely the case the minimum-description-length / identifiability *diagnostic* flags: a value chosen to make one word read out carries no description-length saving that survives the multiplicity of alternatives it was selected from (cf. Section 3; the diagnostic is reported next to every claim, and the operative graduation clause — the order-statistic bar over exactly that multiplicity — is one a hand-fitted value cannot clear). The binding obstacle is not corpus size but identifiability — no known cognate language to constrain the assignment, uncorrected multiplicity, and a sign whose value is chosen rather than anchored.

Read against Section 3, the claim instantiates two distinct failure modes at once. It is **narrative capture** — the table's *structure* says agglutinative and prefixing, but the *label* says Semitic, and the label wins because the desired conclusion (a Semitic prayer) is a fixed prior rather than an emergent reading. And it is a **free-parameter / identifiability** failure — an unanchored sign-value tuned to the target, on a corpus with no known cognate language to constrain it. logos indexes the reading (*301 = /na/) precisely so a model cannot regurgitate it as discovery, and quarantines rather than dismisses it. The lesson generalises: a match rate on a hand-picked word, absent a committed prediction and multiplicity accounting, is not evidence. The *301 claim is the worked example of why the gate must fail closed.

6. Results I: morphology induction

The first worked application of the framework is pre-registered, unsupervised morphology and segmentation induction over the full Linear A corpus (1,341 inscriptions across 52 sites — the 52 that survive logos normalization of Salgarella's 56-findplace envelope). The target claims are the recurring affixes and positional grammar of the Davis/Thomas/Duhoux–Valério tradition: prefixes and suffixes (`i-`, `a-`, `-te/-ti`, and Davis's `i-\*301` verb stem) that field analyses read as genuine Minoan morphology. The pre-registration was frozen before the model touched the data, and every candidate was graded mechanically against two null floors: a within-word sign-order permutation (shuffle) floor, and an `L_fake` floor built from a first-order sign-bigram (Markov) corpus with zero lexical overlap with the real inscriptions. The design earns the paper's core discipline — internal recurrence is not evidence; only separation from a generative null that reproduces sign statistics counts.

Pooled run: an honest null. Of the 16 pre-registered affixes, 4 beat the within-form permutation null and the deflated multiplicity bar (`i-\*301`, `a-`, `i-`, `-te/-ti`), a real confirm rate of 0.250 (Wilson 95% CI [0.10, 0.50]). This clears the shuffle floor decisively: there *is* recurring edge structure, exactly as Davis/Thomas describe. The apparent standout is `i-\*301`, which sits far above the permutation null (six distinct affix environments, $z \approx 8.0$, DSR-corrected $p \approx 1e\text{-}14$) — **but that figure is computed against the weaker within-form permutation null only**, and it does not survive the operative `L_fake` bigram floor. Under the bigram floor the whole set fails: the fabricated Markov corpus confirms the same affixes at rate 0.375 (Wilson 95% CI [0.18, 0.61]), above the real 0.250. On this corpus the apparent affixation — `i-\*301` included — **did not exceed** this first-order bigram control; and because the bigram procedure has **not yet been validated on a known morphologically-structured syllabic corpus** (the Linear-B positive control below), we classify the outcome as **NO POWER**, not as evidence against morphology. `has_morphology_power` is therefore False and the module reports the null. This is a limit of the *data*, not of the field's analysis: Linear A words are short — mean 1.84 signs over all word-*tokens*, median and mode 1, with 76.1% of word-tokens at ≤2 signs (56.5% single-sign) — and on such words "morphology" and "bigram order" are statistically indistinguishable. The premise reconciles with Fuls's 3.3-sign figure once the denominator is matched: Fuls (2015:57, §6.5) computes the mean over a word *list* of "554 complete sign sequences, where duplicates are reduced to one" (GORILA; Godart and Olivier 1985) — i.e. over distinct word *types*, not tokens. Restricting our corpus to distinct multi-sign types likewise recovers 3.07; the token vs. type denominator is the whole discrepancy. We note one tension worth flagging rather than burying: Fuls reads the 3.3-sign length, via his exponent-to-word-length regression, as evidence that Minoan is *polysynthetic* (rich affixation); our direct morphology test finds the apparent affixation bigram-reproducible with no power, so it does **not** corroborate a rich-morphology reading — the word-length typology and the direct null point opposite ways, and we report the null. This bigram free-rider is one of the would-be confounds the machinery has caught and reported as a null rather than a finding. One validation this null still lacks — and the first item of the planned revision — is a **positive control**: metrology and phonology (§7.1–7.2) each demonstrate their detector fires on a planted signal, but the morphology bigram-floor test has not yet been shown to recover *known* affixation, so "no power on Linear A" cannot yet be cleanly separated from "the floor is too strong." The direct control is the identical bigram-floor test on **Mycenaean Greek (Linear B)**, whose longer words carry uncontested inflectional morphology; the DĀMOS Linear-B corpus ingested for §7.3 (13,562 wordforms) now makes it runnable. Our short-word argument makes a falsifiable prediction there — the test *should* regain power on Linear B — which, if borne out, would establish the Linear A result as a short-word / low-power situation rather than a claim that Minoan lacks morphology; until that control is run we report the Linear A outcome strictly as *no power*, not as evidence against affixation.

A modest segmentation positive. The word-boundary segmenter is a genuine, held-out positive. Under leave-one-site-out cross-validation (52 sites; not k-fold, given formulaic dependence), a DP-unigram (Goldwater-style) segmenter recovers the scribe's own word divisions at micro-F1 0.436 versus a random-boundary baseline of 0.389 (boundary base rate 0.406). The gap survives site-level uncertainty: a site-clustered bootstrap (resample the 52 sites with replacement, 4,000 draws) puts the real-minus-random gap at a 95% CI of [0.021, 0.099], excluding zero ($P(\text{gap} > 0) = 0.998$); the paired gap, not the wide marginal per-segmenter CIs, is the operative test because both segmenters are scored on the same resampled sites. Morfessor over-segments the short syllabic corpus (micro-F1 0.156) and is reported as not usable rather than hidden. The *method* works above chance even where the *morphology claim* does not survive the bigram floor.

Stratified re-run. Because pooling the corpus risks inducing "morphology" that is really a mix of genres — a confound the pooling-preserving permutation null cannot catch — a second pre-registered addendum re-ran induction per genre stratum. The strata reconcile exactly to the corpus total: 290 admin + 99 libation + 129 other + 740 seal (structurally excluded as an abbreviation channel) + 83 emptied by abbreviation-strip = 1,341 inscriptions. Induction ran on admin (290 inscriptions, 1,486 words, 14 sites), libation (99, 370, 15), and other (129, 187, 40). Note that the 99-inscription "libation" **genre** stratum — stone-vessel inscriptions over 15 sites — is *not* the same object as the libation-**formula** corpus (~50 inscribed objects drawn from peak sanctuaries *and* caves, across more than five findspots) that carries the recurring dedicatory formula; conflating the two overstates the evidence. Following Salgarella & Judson (2024), chronological cross-validation was explicitly declined (988/1,341 ≈ 74% single-horizon LM IB), and site — not scribal hand, which is not individuated for Linear A (Salgarella 2019) — is the independence unit, gated by presence at ≥2 distinct sites. The candidates are tested for cross-stratum stability above a *per-affix* `L_fake` floor. The result is `has_validated_morphology = False`: no pre-registered affix is cross-stratum stable above the bigram floor at the pre-registered $\alpha = 0.01$. Stratification also showed that even `i-\*301` is bigram-reproducible in the repetitive libation register, where the Markov corpus manufactures the apparent verb morphology; Davis's slot grammar is not refuted, merely not separable from sign bigrams here. Where formula-bearing sites do drive a signal, statistical power is intrinsically low: a leave-one-site-out CV over so few formula findspots (an effective ~5-fold site partition) cannot resolve a modest effect, and we say so rather than read the borderline result as support.

Sensitivity, disclosed. An independent adversarial review corrected an initial overclaim of robustness. The null is a threshold call, and the honest 2×2 is: under both α thresholds at the defensible bucketing (libation = stone vessels only), NULL; under loose bucketing (libation including stone objects and inked), NULL at $\alpha = 0.01$ but at $\alpha = 0.05$ the affixes `i-` (locative) and `-te/-ti` (ablative) validate. These are precisely the Duhoux/Valério H3 nominal affixes that Salgarella (2025) §8 independently endorses — the named borderline candidates, the closest thing to recoverable morphology in the corpus, but fragile to *both* the α threshold and a defensible genre choice. The pre-registered $\alpha = 0.01$ null is itself robust to the bucketing, so the bucketing fix is not cherry-picking the null. Separately, logos's own self-review — not the adversarial reviewer — caught the bigram free-rider `a-` and excluded it via the per-affix floor; the adversarial reviewer also corrected a docstring that mislabelled the ≥2-site presence gate as effective-n deflation. We report the null; full effective-n deflation of the z-test — which would only strengthen it — is a recorded, deferred limitation.

7. Results II: metrology, phonology, and the cross-script asymmetry

Three further pre-registered probes exercise the harness against distinct field claims. Each returns a calibrated null or a truth-capped positive; together they map where the Linear A corpus does and does not carry recoverable signal, and they locate the binding constraint at identifiability rather than effort.

7.1 Metrology: a parse-coverage null that agrees with the field

Direction D poses the accounting tablets as a constraint-optimisation. We parse HT (LM I) documents into balance units of the form `[recipient][commodity logogram][integer][fraction signs]` and solve the fraction-sign rational values by RANSAC over exact `Fraction` Gaussian elimination — robust to the unbalanced-tablet outliers that defeat least-squares — requiring line items to balance the preserved KU-RO totals. Validation is grouped k-fold by document (no tablet straddles train and test) against a permutation null.

On the real corpus (32 documents / 35 balance units), integer-level balance holds for 7/28 (25%). Of seven fraction-bearing constraints, only one is satisfied by the solved system, and the sole value our balance method can determine is the fraction sign J = 1/2, forced by HT104's `2·J == 1`. Held-out fraction balance is 0.0 against a permutation-null mean of 0.113 (max 0.143), $p = 1.0$, not separated: the solved fractions do not generalise above the shuffle null. A synthetic positive control confirms this is not a machinery bug — with planted values the solver recovers all of them and reaches held-out balance 0.9 vs null 0.22, $p = 0.003$, separated. The null is driven by sparse automated parse coverage: only ~7 fraction-bearing constraints survive automated parsing of 32 tablets, because multi-commodity blocks, illegible number glyphs, KI-RO deficit sub-lists, and editorial restorations are not captured.

Crucially, this is a null the field itself reached. Corazza et al. (2021, p.2) abandoned the balance-to-totals method precisely because no Linear A inscription containing numeral-phrase totals is free of reading or calculation problems, moving instead to constraint programming over ordinal/typological premises. logos's parse-coverage null is therefore consistent with the published experts, not a logos-unique failure. The claim is method-specific: our balance method determines only J = 1/2 — but that value is not novel, it is the long-standing consensus (Bennett 1950), corroborated here independently by Corazza et al. (2021) and by Schrijver (2014). Schrijver's three further J = 1/2 tablets (HT Zd 156, PE 1, HT 123a) lift the value to at least four corroborating documents. The result illustrates the harness working as designed: a claim that fails held-out validation is reported as a null, and the one value that survives is credited to its actual originator rather than re-branded as a logos discovery.

7.2 Distributional phonology: a data-limited null bounded by a power curve

Track 2 asks whether a sign's distributional context — which signs sit beside it — carries information about its phonetic value, measured on the known `AB` signs whose Linear B sound values can be borrowed. Features are PPMI-weighted, L2-normalised left/right co-occurrence counts; labels are the Linear-B-transferred (C, V) parsed from the GORILA name, so features never see the labels. The test is leave-one-out 1-NN against a label-permutation null, Šidák-deflated over the two tests; 50 of ~55 AB signs met the ≥3-occurrence threshold.

Neither test clears chance: consonant class (13 classes) at 0.080 accuracy (null 0.065, permutation $p = 0.40$) and vowel class (5 classes) at 0.140 (null 0.201, permutation $p = 0.86$). The honest verdict is *data-limited*, not *no bridge exists*, because the positive control disciplines the reading. A first draft's flat positive control failed to fire at planted strength 1.0; replacing it with a power curve over planted strengths shows the test firing ($p < .05$) at strength ≥ 2 but not at 1. The test works but is not infinitely sensitive at n = 50 / 13 classes, so we cannot distinguish "no distribution→phonetics bridge" from "too few signs to detect a weak one." Both point the same way: the sound path is not recoverable from Linear A alone. The obvious next move — impute sound values cross-script from a larger Linear B corpus — is now *tested rather than deferred*: §7.3 reports that ingesting the full DĀMOS Mycenaean corpus leaves the cross-script distributional alignment at chance, so more Linear B data is necessary but not sufficient; the block is an absent distributional signal, not corpus volume alone. No sound value is imputed for any sign. This pre-empts the predictable "couldn't you just use ML to recover the sounds?" objection with a quantified detectability floor rather than hand-waving.

7.3 The cross-script asymmetry: image alignment recovers shared anchors, but the recovery is circular by construction

The palaeographic probe renders 88 Linear-A and 74 Linear-B glyphs and tests held-out recovery of shared A↔B anchors. Classical descriptors (HOG + Hu moments + shape-context) reach direct-NN accuracy 0.410 [0.18, 0.64] and Procrustes 0.185 [0.00, 0.36], far above image chance (0.0135); a from-scratch I-JEPA reaches only 0.324 ± 0.025 (3 seeds) and is excluded from the claims as over-parameterized for ~90 glyphs. Image embeddings recover held-out anchors where sequence co-occurrence sat at chance. Two controls decide how to read that asymmetry, and together they *downgrade* it from a positive to a demonstration.

First, a font-swap control. Rendering both scripts in one typeface (Aegean, George Douros) risks making the alignment a single-designer house-style artifact rather than a fact about the signs. We re-render Linear A in Noto Sans Linear A and Linear B in Noto Sans Linear B — two independent type designers, each font covering only its own Unicode block (Noto-A carries zero Linear B glyphs and vice-versa), so no single hand draws both scripts. The recovery survives and is if anything stronger (direct-NN 0.367→0.416, Procrustes 0.156→0.195 on the control's matched anchor set; both ≫ chance): the font-artifact hypothesis is *refuted*, and "distances reflect shape, not font" is now demonstrated rather than asserted (`scripts/palaeo/font_control.py`).

Second, and decisively, the circularity the font control cannot remove. Linear B was historically adapted from Linear A, and the A↔B transcription *values* that define the anchor set were assigned by twentieth-century epigraphers largely on the basis of sign-shape similarity. An image encoder that "recovers the A→B value map" therefore re-derives a correspondence that was *defined* by shape — the result is close to tautological and is not independent evidence for any phonetic reading. We accordingly relabel this leg a **shape-genealogy engineering demonstration, not an offensive positive**: it confirms a known palaeographic fact (borrowed signs look alike) and validates a font-robust descriptor pipeline, but it reads no Minoan. It remains truth-layer-capped in any case — the Ferrara, Montecchi & Valério "Archanes Formula" hard negative (`CH 095` resembles `AB 60`/*ra* but is not its phonetic counterpart) shows a plausible shape match can be phonetically wrong, so a shape match is a ≤ 0.75 signal (invariant 5), never a reading. The honest asymmetry is thus: the image path re-derives a shape-defined mapping (expected, circular, capped), while the sequence and cognate paths **remain at chance even after ingesting the full DĀMOS Mycenaean corpus** — 13,562 wordforms, ≈15× the earlier cognate file, 56 anchors: the best of five alignment methods reaches 0.025 against a 0.011 chance floor (`clearly_above_chance = False`), while the *positive control recovers 0.947*, so the harness works and the null is a real null. Ingesting all of DĀMOS therefore reframes the obstacle from *data volume* to *absent distributional signal*: the A↔B phonetic correspondence is simply not carried by cross-script sign co-occurrence, even at full Linear-B scale. And the A-only signs on which offensive claims would pivot (e.g. *301) have no cross-script anchor at all, so images cannot impute them even in principle. **The study reports no offensive positive.**

8. Results III: contamination and information-sufficiency probes

Two failure modes are invisible to the false-positive / multiplicity axis of §7: a decipherment that *looks* novel but silently reproduces published readings a language model has memorised, and a corpus that may be too small — or too unidentifiable — to recover any reading at all. This section reports the decontamination and information-sufficiency probes that target them. Both are run as method demonstrations. The LLM-ablation and `L_virgin` probes return completed (if deliberately bounded) findings; the information-sufficiency sweep is reported as a known-answer upper-bound protocol, with the accuracy-vs-size curve deposited in the repository (§8.3). Throughout, where a clean number was not obtainable we report the partial-null and the confound rather than a headline.

8.1 LLM-ablation: does a frontier model regurgitate the literature?

We ran the §C.4 ablation against `qwen2.5:72b` — a frontier-scale open model, a far stronger memoriser than a local 12B — on a rented H100 (40 seeds × 40 held-out Linear A forms; NW-Semitic candidate; a Hebrew lexicon of 7,682 skeletons for the model-free arm; 0 LLM errors across 40/40 seeds) (logos v0.1.0). The first-pass contamination rate came out at `1.000` in all 33 seeds where it was defined — but this is an **artifact, reported as such**: the mechanical arm never shares a literature-matching correspondence with the LLM (their value vocabularies overlap in only 10 of 134 values), so the rate is forced to 1.0 by construction. Publishing it as a contamination result would itself be the overfitting failure the discipline layer exists to prevent (logos v0.1.0).

The gate-2 rerun, reconciled to a common consonant-skeleton vocabulary, yields the honest deliverable: a per-word reproduction table over 40 seeds. `qwen2.5:72b` reproduces Gordon's `JA-NE` = *yn* / "wine" in 14/40 seeds and Best 1981's `A-SA-SA-RA-ME` = "oh Asherah!" in 2/40 — published readings the model-free edit-distance baseline never independently reaches (0/40) — while never reproducing the obscurer vessel equations `KA-RO-PA` and `SU-PA-RA` (0/40) (logos v0.1.0). Three of the four tested lexical items — `JA-NE`, `KA-RO-PA`, `SU-PA-RA` — are among Gordon's five West-Semitic equations (as is the separately-treated `KU-RO` below); the fourth, `A-SA-SA-RA-ME`, is Best's, not Gordon's. An external red-team correction split `KU-RO`: of its 14/40 hits, 8/40 route through the Semitic root *kull* and 7/40 are reachable via the general-knowledge meaning "total" (confirmed by tablet arithmetic; the two categories overlap on one seed), so `KU-RO` = "total" is **excluded** as a witness and `KU-RO` = *kull* is downweighted (logos v0.1.0). The surviving gradient — famous readings reproduced, obscure ones not — is what C.4 detects: **reproduction of published readings weighted by web-prevalence**, not verified correctness. The gate-2 rate itself (mean 0.646) remains confounded by the baseline's representational ceiling and is not reported as a contamination measure.

8.2 L_virgin: memorisation vs. reasoning

The §C.2 `L_virgin` probe shows a sign in real context and asks whether the model's behaviour on *published* signs generalises to signs **unseen in the indexed literature** — absent from the indexed sources, hence not memorisable from them (never "impossible to have seen" in any absolute sense) (logos v0.1.0). Across 40 seeds, `qwen2.5:72b` assigned a value to the 20 known AB signs in ~20/40 seeds but to the 11 virgin *-series signs in only ~1/40 — it **abstains ~97% of the time** on the untouched signs while confidently valuing the memorisable ones. The apparent known−virgin consistency gap (0.601, permutation $p = 0.0005$) is coverage-confounded and was flagged, not asserted: the matched-*n* robustness check returned `no_power` (0 of 11 virgin signs reached $n \geq 10$), so the rigorous comparison could not run. The interpretable signal is the abstention asymmetry itself, consistent with regurgitation-dominated behaviour — and, per the module's standing rule, **consistency is not correctness**: with no ground truth on the virgin signs, this bounds how the model behaves, never whether it is right (logos v0.1.0). The full arc cost ≈ $2.80 in ephemeral GPU rental.

8.3 Information-sufficiency upper bound: a known-answer CSA protocol

The learning-curve harness reframes the withdrawn unicity-distance diagnostic — a toy-model precondition, never a sufficiency or refutation claim — as a sharper, answerable question: *is a Linear-A-scale corpus (539 distinct word-forms; $V = 259$ sign-types at $N \approx 5{,}792$ parsed syllable-signs) large enough to recover a decipherment IF a known cognate existed?* We report the design and its reading rules here as a known-answer upper-bound **protocol**; the full accuracy-vs-size curve, with per-size permutation floors, is deposited in and completes in the repository, and this paper asserts no interim result from it.

The protocol downsamples known-answer benchmarks — Linear B / Mycenaean Greek as the primary syllabic analog, sourced through DĀMOS, and further syllabic corpora — and measures held-out cognate-ID recovery against a permutation chance floor, using a published cognate-ID matcher (Tamburini 2025) that reports the Luo et al. 2019 accuracy metric (logos v0.1.0; [repository URL withheld for review]). Because every benchmark is handed a real cognate signal Linear A lacks, any recovery curve is an **optimistic benchmark, conditional on a known cognate relationship** — a best case Linear A does not enjoy — readable two ways: at-floor at Linear-A scale ⇒ information-insufficient even given a cognate; above-chance ⇒ the binding constraint is identifiability — cognate-absence and uncorrected multiplicity over a mixed sign inventory — not corpus size. Neither branch is ever a decipherability claim, and neither asserts "there is enough text." Using a data-size recovery curve to bound decipherability is not itself new — decipherment learning curves over cipher length (Ravi and Knight 2008; Berg-Kirkpatrick and Klein 2013), the sample-size dependence of automatic cognate detection (List 2014), and analytic corpus-sufficiency thresholds (Barber 1974; Zhang and Gong 2016) are the ancestors; our contribution is the specific repurposing to a cross-language cognate-ID sufficiency locator for an undeciphered target, not the learning-curve idea.

The matcher is the parallel coupled-annealing path validated in the artifact against its published known-answer benchmarks (a correctness check that the reproduction reaches the published Table-3 values); the accuracy-vs-size curve itself — downsampling each benchmark and reading its recovery against the per-size permutation floor under the two-way logic above — is computed and deposited in the repository, and this paper draws no Linear-A-scale conclusion from it. The phonological / sound path, by contrast, is no longer data-deferred: the

DĀMOS Mycenaean holdings are now harvested and ingested, and §7.3 reports that even the full corpus leaves the cross-script distributional alignment at chance — so that bound is a property of the signal, not a promissory note against data yet to arrive. We publish the harness and its bounding logic ([artifact DOI withheld for review]).

9. Discussion, limitations, and conclusion

9.1 The discipline is the contribution

The deliverable of this work is not a reading of Linear A. It is a *harness* — a literature index with a known-reading versus literature-unseen partition, a fabricated-language control corpus, model-ablation contamination checks, multiple-comparison correction, and pre-registration, all graded mechanically from persisted artifacts — together with a body of pre-registered, calibrated nulls, each with its borderline case named. The stratified morphology re-run is the template: at the pre-registered $\alpha = 0.01$, no pre-registered affix is cross-stratum stable above the bigram floor, robust to the genre-bucketing choice; the closest candidates are the Duhoux/Valério H3 nominal affixes (`i-` "to/at", `-te`/`-ti` "from/of"), which surface only in the doubly-relaxed corner ($\alpha = 0.05$ *and* loose bucketing) (logos v0.1.0). A null that names its threshold, its sensitivity, and its near-miss is a stronger referee object than a flat negative.

The axis this harness targets is the **false-positive / multiplicity axis** — the mechanism by which a determined analyst "translates" almost anything on a small corpus. As set out in the Introduction, this is a property of inference methodology and epistemics, not of the object or its data; it is complementary to, not a member of, the fifteen data-and-object obstacles enumerated by Braović et al. (2024), and distinct from the evaluation problem their §6.3 gestures at.

9.2 Two named failure modes for the methods literature

(i) Narrative capture. Structure says X; the label says the desired Y; the label wins. In the June-2026 *301 claim the published table segments an agglutinative, prefix-stacking verb, yet the assigned language family is Semitic and the desired output ("prayers to a Goddess in a matriarchal society") operates as a fixed prior, so the conclusion conforms to the prior rather than emerging from constrained readings. This is distinct from statistical overfitting and must be named separately. Notably, the current synthesis independently concludes Minoan is agglutinative and a language isolate (Salgarella 2025 §8), which the segmentation matches while the Semitic label does not — the mismatch is empirical, between the label and the primary descriptions of the form, not an a-priori typological impossibility.

(ii) Free parameters exceeding the information floor. A lexicon reported as "408 terms and counting" on a fixed corpus has more degrees of freedom than the data can identify; a decipherment whose lexicon grows weekly is underdetermined by construction. The minimum-description-length / unicity *diagnostic* (`k ≤ U_floor`) is built to flag exactly this — after pre-submission review it is **reported next to every claim rather than enforced as a graduation clause** (§3), because clearing it does not certify identifiability: the single sign the whole reading pivots on carries no cross-script anchor (its AB-number and Linear B columns are empty), so its value was assigned to fit the desired root — the textbook free-parameter case the machinery flags as underdetermined. The refutation rests on the claim's own published table, archived in the artifact (version and details withheld for review).

9.3 The inverted pipeline

These failure modes trace to sequencing. The *301 claim publicly commits to an inverted pipeline: declare "officially deciphered," then vet later. logos's documented inverse is **pre-register → held-out → locked artifact**: a hypothesis is hash-keyed and timestamped before it is tested, graded mechanically against held-out material, and the artifact is fixed before any claim. The libation-formula corpus (~50 inscribed objects over >5 findspots, spanning peak sanctuaries and caves) offers a natural held-out cross-validation, but a leave-one-site-out split over so few formula sites has low statistical power — and site-level CV is in any case necessary, not sufficient, since within-site scribal-hand correlation must also be controlled (Salgarella & Judson 2024). This formula corpus should not be conflated with the 99 stone-vessel inscriptions of the "libation" genre stratum (§6): they are different objects.

9.4 Identifiability, not corpus size, is the binding constraint

We take care not to write a defeatist framing, and equally not to over-claim from corpus size. On the real corpus (1,341 inscriptions; N ≈ 5,792 parsed syllable-signs, V = 259) the toy-model unicity distance is U ≈ 204–415 signs — a *precondition* diagnostic only. We explicitly withdraw any reading of this figure as a sufficiency or refutation claim: a lossy, non-injective sign→sound channel may have no unicity distance at all, so "$U \ll N$" does **not** license the statement that there is enough text to pin a decipherment (this is the stance of §4, and §9 now aligns with it). The binding constraint is *identifiability*: the absence of a known cognate language to map to, an uncorrected multiple-testing search burden, and a mixed sign inventory. (Note that the ~7,400 sign-occurrence figure sometimes quoted — Schoep 2002; Salgarella 2025 §1–3 — counts a different channel than the 5,792 parsed syllable-signs used for U here.) Salgarella (2025 §8) reaches the field-level version of this by a different route: the etymological method is "fraught with problems," vocabulary comparison is "not probative," and she names digital/statistical analysis as "the next desideratum in the field."

9.5 Study-wide error budget

Multiplicity discipline operates at two levels, and we are explicit about both. Within a single probe, matches are deflated for the signs × roots × hypotheses searched (the effective-n machinery of §3–4). Across the study, the unit of multiplicity is the **per-probe pre-registration**: the **five executed probes** (morphology, metrology, phonology, cross-script image alignment, and contamination) each carry their own pre-committed α, and every probe is reported regardless of outcome. (The sufficiency sweep of §8.3 is a known-answer *upper-bound* diagnostic on benchmarks handed a real cognate signal — not a Linear A probe — so it is not among these five; on the corrected matcher its accuracy-vs-size curve is computed and deposited in the repository (§8.3), and any above-chance recovery on known-answer benchmarks is true by construction, never a Linear A positive requiring multiplicity protection.) There is no selective reporting: the one probe that returns an above-chance number — the cross-script image leg (§7.3) — is reported as a circular *engineering demonstration*, not a positive. On the battery-level correction the paper's own rule demands: four of the five executed probes are nulls sitting far from any threshold (metrology p = 1.0; phonology p = 0.40 / 0.86; morphology at no power; contamination an artifact by construction), so a family-wise/FDR correction over the probe-level tests is arithmetically trivial and only makes each *more* null — and the fifth probe surfaces **no positive for such a correction to protect** (the image leg is capped at ≤ 0.75 and relabeled circular). Battery-level multiplicity is therefore not load-bearing here: there is no surviving positive that a correction could threaten. Nothing in the battery is read as a decipherment.

9.6 Limitations

The nulls are honestly bounded, not decisive. Full effective-n deflation of the morphology z-test is deferred; the current control is a conservative ≥ 2-distinct-sites presence gate, which would only strengthen the null (logos v0.1.0). No held-out *decipherment* has been achieved, and the study reports **no offensive positive** — the deliverable is calibrated absence, not a reading. The one probe that recovers an above-chance number, cross-script sign-image alignment (direct-NN 0.410 [0.18, 0.64]; Procrustes 0.185 [0.00, 0.36]), is a font-robust but circular *engineering demonstration* (§7.3): a font-swap control refutes the typeface-artifact worry, but the A↔B anchor values were themselves assigned by sign shape, so the recovery re-derives a shape-defined mapping and is truth-capped at ≤ 0.75, never a reading. No phonetic claim is made. The philological points in §5 rest on the published literature (Davis 2013, Duhoux 1992, Thomas 2020, Salgarella 2025, Steele & Meissner 2017) rather than on independent Aegean-epigraphic review, which we disclose as a limitation; because those points are cited rather than original, the argument does not depend on any new sign reading of our own. Chronological cross-validation is largely foreclosed because roughly 74% of the corpus sits in the single LM IB horizon, leaving site-held-out splits as the only realistic partition. And the corpus itself — small, partly lossy, with no bilingual yet found (Salgarella 2025 §8) — sets the ceiling.

9.7 Conclusion and reproducibility

If a Linear A decipherment is attainable on present evidence, this framework is built to recognize it; but the contribution is the framework and the calibrated null itself — the instrument and the honest limits of what the corpus can support — not a reading, and the calibrated null is the field's insurance policy against another false one. The code, pre-registrations, and all findings are open under the MIT licence and archived on Zenodo ([artifact DOI withheld for review]; repository [repository URL withheld for review]). The licensed epigraphic corpora — GORILA-, SigLA-, lineara.xyz-, and DĀMOS-derived — are not redistributed, per their CC BY-NC-SA / all-rights-reserved terms; each source's licence and retrieval path is documented (data-provenance), so the harness is reproducible against a locally obtained corpus. This work is independent and not affiliated with, or endorsed by, any of the cited projects or their authors. ## References

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